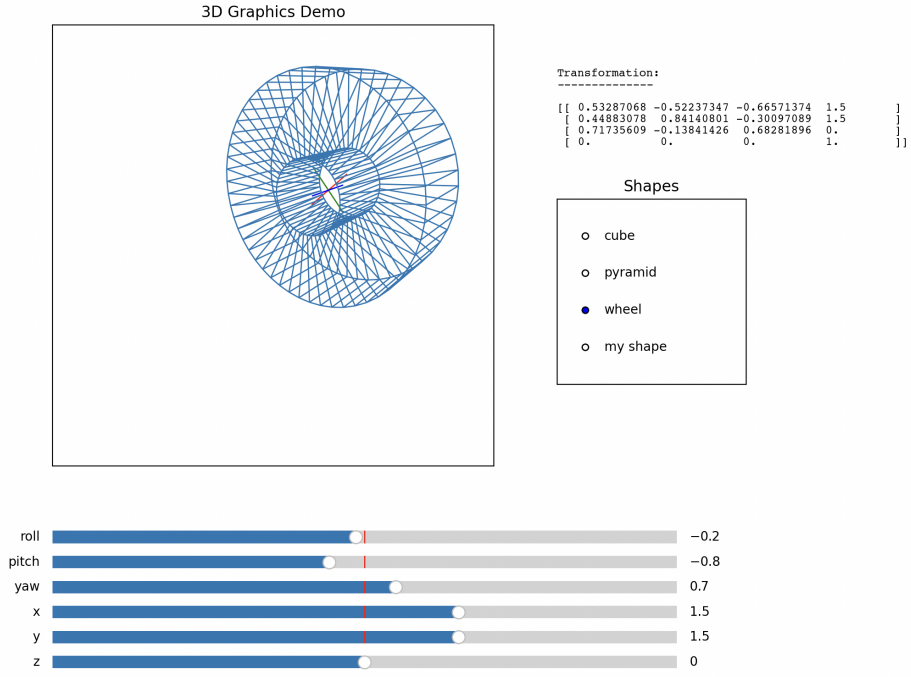
# Lab 4: Linear Transformations and Graphics

`CAS CS 132`: *Geometric Algorithms*

Due November 12, 2026 by 8:00PM

**Outline.** The objective of this lab is to build an interactive `matplotlib` widget that manipulates 2D renderings of 3D wireframes. Once you're done, the widget should look something like this.




You won't be required to submit a lab write-up. Instead, you'll submit your code to Gradescope and it'll be autograded. There is a very small extra credit part that can be optionally submitted as the write-up.

## The Widget

The starter code is given in `lab4.zip` and has two files, one called `lab4.py` that you'll edit, and the other called `demo.py` that contains the code for the widget itself. Don't change the names of these files or the names of any functions

included in the starter code. The only changes you should make are to fill in the provided TODO items. At a high level, the code works as follows:

1. 3D wireframes are represented by 3D line segments. 3D line segments are represented as pairs of 3D points. 3D points are represented as 3-tuples of floating-point numbers. This means we represent 3D wireframes as lists of pairs of 3-tuples of floating-point numbers. The function `shape_to_matrix` takes a 3D wireframe with $n$ line segments and converts it into a $4 \times 2n$ matrix in which each column is an endpoint of a line segment in the wireframe in *homogeneous coordinates*.

2. The output of `shape_to_matrix` will be transformed by the output of `transform_matrix`, which should implement a combination of rotation and translation. See the docstring for details. **Important:** Translation should happen *after* rotation.

3. 2D line segments are represented as pairs of 2D points, which are represented as pairs of floating-point numbers. The function `matrix_to_shape` converts a $4 \times 2n$ matrix into a list of $n$ 2D line segments that will be rendered on the screen. This is done by

   (a) applying the projection matrix with viewing position `(0, 0, 10)` to the given matrix,

   (b) then converting each column from homogeneous coordinates to 2D Cartesian coordinates via the transformation discussed in lecture and lab,

   (c) then grouping every *two* points to create a list of line segments.

4. The above steps are combined into a single function called `full_transform`. This function should:

   (a) convert the 3D shape to a matrix,

   (b) apply the transformation matrix, and

   (c) convert the matrix to a 2D rendering.

Once you've completed every TODO item, you should be able to run `python3 demo.py` to see the widget in action.

## Extra Credit

For a small amount of extra credit, you can build a reasonably complex wireframe[1] using the variable `extra_credit_shape` which can be viewed in the widget. In order to get credit, you must submit a PDF write-up that includes the following.

[1] We have final say on what "reasonably complex" means.

1. A *high-quality* screenshot of the demo with your shape in a non-default orientation. If the image is small, blurry, a photo of your computer screen, doesn't contain the entire demo screen, or anything of this nature, you will receive no credit.

2. You must name your shape, using the variable `extra_credit_name`. If your picture says "my shape" you will receive no credit.

3. One or two sentences about how you generated your wireframe, e.g., whether you did it programmatically or by hand, where you got the inspiration, etc.

4. You must clearly state in your write-up if you're comfortable making the image public (in case we collect the results on a public-facing website). If you do not explicitly state this, you will receive no credit.

## Submission

This week, you'll upload the single file `lab4.py` to Gradescope under the assignment **Lab 4 (Programming)**, where you can verify that it passes a few (but not all) autograder tests. **You must pass all tests to receive credit for this lab.** Test your system early; there may be system-dependent issues that we'd like to address as early as possible. For the extra credit, you'll submit a PDF write-up (as you did in previous weeks) under the assignment **Lab 4 (Extra Credit)**.